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COURSE CODE : CSA613

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**COURSE NAME :Design and
Analysis of Algorithms**

C01-ASSESSMENT TOOL 03 -TECHNICAL PRESENTATION

**TITLE : Asymptotic Analysis of Sorting Algorithms Analyze the efficiency of Merge
Sort and Quick Sort using asymptotic notations.**



PROJECT DESCRIPTION

This project presents a detailed analysis of the efficiency of Merge Sort and Quick Sort using asymptotic notations such as Big-O (O), Big-Theta (Θ), and Big-Omega (Ω). Both algorithms are based on the Divide and Conquer technique, but they differ in their execution and performance under different conditions. The project evaluates their best-case, average-case, and worst-case time complexities, space requirements, stability, and practical performance on different input sizes. Merge Sort guarantees $O(n \log n)$ time complexity in all cases but requires additional memory, whereas Quick Sort is generally faster in practice with an average time complexity of $O(n \log n)$ but may degrade to $O(n^2)$ in the worst case due to poor pivot selection. By comparing these algorithms through theoretical analysis and performance evaluation, the project helps understand their advantages, limitations, and suitable applications, enabling the selection of the most efficient sorting algorithm for different computational problems.